

Aryan Malyala

3rd Year Undergraduate

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Academic Qualifications

Year	Degree/Certificate	Institute	CGPA/%
2028	B.Tech	Indian Institute of Technology, Kharagpur	6.82/10
2024	Class XII	Narayana Junior College, Pati, Telangana	96.1%
2022	Class X	St. Anthony's School, Sangareddy, Telangana	9.7/10

Projects

InterviewForge

Python, Gemini API, JSON | Self Project

May 2026 - Jun 2026

- Conducted **personalized** interviews by evaluating candidate responses and adapting interview flows based on performance.
- Generated **adaptive interview questions** tailored to candidate skills and previous responses across multiple technical domains.
- Analyzed candidate strengths and weaknesses to deliver real-time feedback with targeted recommendations through **Json**.
- Managed resilient API communication through **exception handling** and automatic API key rotation under API request rate limits.

Coding Question Search Engine

Python | Self Project

Jun 2026

- Developed a coding question search engine using **TF-IDF**, inverted indexing, and cosine similarity for efficient document retrieval.
- Implemented an **inverted indexing pipeline** with TF-IDF, tokenization, and document indexing for efficient keyword retrieval.
- Optimized **cosine similarity-based** query processing and retrieval performance using efficient data structures across problems.

Sorting Visualiser

C++, SDL2 | Self Project

May 2026

- Built an interactive visualizer for **Bubble, Selection, Insertion, Merge, and Quick Sort** using smooth SDL2 animations efficiently.
- Developed controls for **algorithm selection, array size, and visualization speed** for enhanced user interaction and better usability.
- Designed a scalable **object-oriented architecture** using inheritance and polymorphism to support multiple sorting algorithms.

Rubik's Cube Solver

C++ | Self Project

May 2026

- Developed a C++ Rubik's Cube solver using advanced graph search algorithms for efficient and optimal state space traversal.
- Built a efficient reverse BFS database of over **109M cube states** and implemented **DFS, IDDFS, and IDA*** with state encoding.
- Optimized search using heuristic-guided pruning and move generation, reducing explored states while preserving optimality.

Awards and Achievements

- Solved **600+ algorithmic programming problems** across leading competitive programming platforms with consistency
- Achieved a peak **LeetCode rating of 1621** and attained **Pupil** on **Codeforces** with a peak rating of **1240**
- Secured an All India Rank of **2391** in **JEE Advanced 2024** among over **1.80 lakh** candidates nationwide.
- Secured an All India Rank of **9025** in **JEE Main 2024** among over **14 lakh** candidates nationwide.
- Secured an All India Rank of **39** in **Indian Statistical Institute Examination 2024** among over 50,000 candidates.
- Achieved a Rank of **677**, placing in the top **0.21%** of candidates in **EAMCET 2024** administered by the Telangana State

Skills And Expertise

- Programming:** C | C++ | Python
- Libraries:** C++ STL | SDL2
- Core:** Data Structures and Algorithms | Object-Oriented Programming
- Tools:** Git | VS Code

Coursework Information

- Programming / Mathematics:** Programming and Data Structures* | Advanced Calculus | Linear Algebra and Optimisation Models | Linear Algebra: Numerical and Complex Analysis | Probability and Stochastic Processes
- Electronics:** Analog Electronic Circuits* | Network Theory* | Digital Electronic Circuits* | Signals and Systems* | Systems and Control | Digital Signal Processing-1 | Analog Communication

*Includes Laboratory Component

Positions of Responsibility & Extra Curriculars

- Contributed to community development initiatives during a **National Service Scheme(NSS) camp** through active participation in social outreach and volunteer programs
- Active member of **Prasthanam (Technology Dramatics Society)** — performed in stage productions, contributed to scriptwriting and event promotions